

Understanding the GRUB Bootloader



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Overview

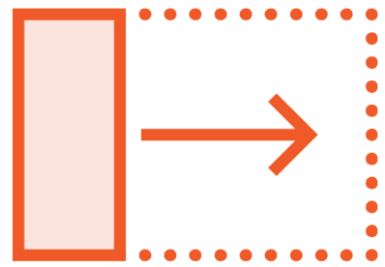


GRUB - Grand Unified Bootloader

- Booting Linux
- Initramfs
- Kernel
- Kernel Parameters
- GRUB Configuration
- Reinstalling GRUB
- Recovery



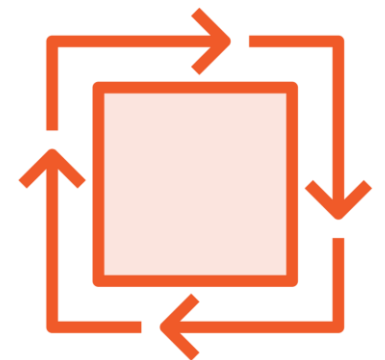
Booting Linux



BIOS: Looks for bootable disk



GRUB Menu is displayed



GRUB loads the selected Kernel and Initramfs



```
$ sudo vim /etc/default/grub  
$ sudo vim /etc/default/grub.d/50-cloudimg-settings.cfg  
$ sudo update-grub
```

Configure GRUB to Display

We must first configure GRUB to display the menu. This is normally via the `/etc/default/grub` file but we also must check for settings being reset in `/etc/default/grub.d/`

Demo



Configure GRUB:

- Set timeout
- Display menu



Loading Linux

Kernel

The core OS but with many device drivers that need to be loaded from `/usr/lib` or `/usr/lib64`

Initramfs

Has drivers to allow the Kernel access to the root filesystem, is only required for the initial startup



Demo



Let's investigate boot Linux:

- Kernel
- Initramfs
- /boot



```
init=/bin/bash  
# passwd  
# exec /sbin/init
```

Kernel Parameters

We can pass parameters to the Kernel that can help us recover from issues. For example, if we have not set the root password and sudo has become unusable we are able to reset or recover the root user's password.

Demo



Kernel Parameters:

- `init=/bin/bash`
- Recover root password



```
$ sudo nano /etc/grub.d/40_custom  
$ sudo update-grub
```

Custom GRUB Entries

We can add custom GRUB entries to the 40_custom file. We can use the editor that we are most comfortable with. We can copy entries from /boot/grub/grub.cfg as a starting point

Built-in Recovery Options

Using the advanced sub-menu, we can boot to the the recovery menu that has some repair tools available including the root user shell

```
Recovery Menu (filesystem state: read/write)
resume      Resume normal boot
clean       Try to make free space
dpkg        Repair broken packages
fsck        Check all file systems
grub        Update grub bootloader
network     Enable networking
root        Drop to root shell prompt
system-summary System summary
```

<Ok>



Demo



Investigating Built-in Repair:

- Menu options



```
Boot server install ISO
Spacebar to see menu
F6 Other options
# mount /dev/sda1 /mnt
# mount --rbind /dev /mnt/dev
# mount --rbind /proc /mnt/proc
# mount --rbind /sys /mnt/sys
# chroot /mnt bash
# grub-install /dev/sda
```

Recovery ISO

If GRUB itself has become corrupt, then it needs to be reinstalled. We can download the ISO file and boot to this to get an OS to run.

Demo



Boot to live ISO:

- mount root
- chroot
- repair grub



Summary



Booting Linux and Grub

- /boot
- initramfs
- kernel
- kernel parameters init=/bin/bash
- root password recovery
- reinstalling grub
- custom grub entries
- update-grub



Up Next:

Managing Software in Ubuntu 20.04

